barely alter behaviour. The example's `1e-4` is a starting point for adapters, not a universal optimum.

Epochs specify passes over the training set. Two epochs mean each example is encountered twice, subject to training details. More passes are not automatically better: small datasets can be memorised quickly. Batch size controls how many examples fit in each forward pass; gradient accumulation combines gradients across several passes before an optimiser update, approximating a larger effective batch without holding every example simultaneously in memory.

Sequence length is the number of model tokens retained for a training sequence. Increase it only when examples genuinely require more context. Long sequences consume more memory, and truncation can silently remove the completion you hoped to teach. Inspect tokenised examples before launching a large run.

LoRA rank, written r, controls the capacity of the low-rank update. Higher rank usually means more trainable parameters, memory and capacity, not guaranteed quality. Alpha scales the adapter contribution; dropout provides regularisation. Target modules decide which neural-network layers receive adapters. `all-linear` is an established PEFT option, though architecture and resources may justify a narrower selection. Change one setting at a time and log the result.

When four-bit QLoRA makes sense

When the unquantised base is too large for your available VRAM, QLoRA may help. Transformers integrates BitsAndBytesConfig with four-bit NormalFloat, or NF4, quantisation. The Hugging Face documentation recommends NF4 for training four-bit base models and supports using bfloat16 computation where the accelerator permits it.

A minimal configuration is `BitsAndBytesConfig(load_in_4bit=True, bnb_4bit_quant_type="nf4")`. Pass this quantisation configuration to a supported model-loading or trainer path while retaining the PEFT adapter. For a practical build, also check the current PEFT guide on preparing quantised models for k-bit training; details vary by library version and training backend. Quantised base weights stay frozen: you train the extra parameters, not arbitrary four-bit weights directly.

Remember the wider hardware bill. GPU memory holds more than the model weights: activations, attention calculations and temporary buffers matter. An impressive claim that a large model "fits in 8GB" may refer to inference, a short training sequence or a carefully optimised configuration. Do a short trial before leaving a full training job unattended.

If configuring CUDA or BitsAndBytes becomes more work than the actual experiment, consider Unsloth's maintained

